

FarmerChat Mobile Automation -- Stakeholder Report

Project: FarmerChat (org.digitalgreen.farmer.chat)

App Version: 4.0.0

Date: March 2026

Prepared by: Shaik Mohamed Imran

1. Executive Summary

A comprehensive end-to-end UI automation test suite has been built for the FarmerChat Android application using the **Maestro** framework. The suite covers **40 test cases** across 8 functional modules, running on both local emulators (via Jenkins CI/CD) and **BrowserStack real devices** (Samsung Galaxy S24, Android 14).

Metric	Value
Total Test Cases Authored	40
BrowserStack-Compatible Tests	35
Reusable Helper Flows	9
Orchestration & Reporting Scripts	8
BrowserStack Real Device Pass Rate	91% (32/35)
Target Device (Cloud)	Samsung Galaxy S24 -- Android 14
Target Device (Local)	Pixel 7 Pro Emulator -- API 34
CI/CD Platform (Local)	Jenkins
CI/CD Platform (Cloud)	BrowserStack App Automate

2. Functional Coverage

2.1 Module Breakdown

Module	Test Cases	Priority Split	Coverage
Onboarding	TC01, TC02, TC03, TC04, TC31, TC32, TC37	2x P0, 1x P1, 4x P2	Language selection, name entry, skip flows, legal links
Home Feed	TC05, TC07, TC08, TC09, TC10, TC12	2x P0, 4x P1	Weather widget, feed scroll, Learn More, gender/crop cards
Chat / AI	TC06, TC11, TC13, TC14, TC24, TC26, TC28, TC30, TC34, TC35	5x P0, 5x P1	Type question, AI response, share, save, listen, follow-up, gallery/camera upload, close chat
Settings	TC15, TC16, TC18, TC27	2x P0, 1x P1, 1x P2	Display mode, update name, skip signup, logout
Authentication	TC17, TC19, TC25	3x P0	Phone signup (settings & drawer), login with OTP
Help & Support	TC20, TC21, TC22	3x P2	FAQ accordion, Terms of Use, Privacy Policy
Navigation	TC23, TC29, TC33, TC36	2x P1, 2x P2	Language switch, location skip, recent chats, drawer home
Error Handling	TC38, TC39, TC40	2x P0, 1x P1	Airplane mode recovery, chat retry, history retry

2.2 Priority Distribution

Priority	Count	Description
P0 (Critical)	14	Core user journeys -- onboarding, chat, auth, logout, error recovery
P1 (High)	16	Important feature validations -- feed content, sharing, settings
P2 (Medium)	10	Supporting features -- help pages, legal dialogs, display mode

3. Test Architecture

3.1 Framework & Tools

Component	Technology
Test Framework	Maestro (mobile UI automation)
App Under Test	FarmerChat Android (org.digitalgreen.farmer.chat)
Local Execution	Android Emulator (Pixel 7 Pro, API 34)
Cloud Execution	BrowserStack App Automate (Samsung Galaxy S24, Android 14)
CI/CD (Local)	Jenkins with automated HTML report generation
CI/CD (Cloud)	BrowserStack Maestro API v2
Reporting	Custom HTML reports with video recordings, execution logs, priority badges

3.2 Project Structure

```
Maestro_automation_FarmerChat/
├── flows/
│   ├── onboarding/           # 4 test flows (TC01-TC04)
│   └── home/                 # 41 test flows (TC05-TC40 + sub-flows)
├── helpers/                  # 9 reusable sub-flows
│   ├── complete_onboarding.yaml
│   ├── navigate_to_settings.yaml
│   ├── navigate_to_help.yaml
│   ├── navigate_to_chat_history.yaml
│   ├── open_drawer.yaml
│   ├── weather_share_location_and_back_home.yaml
│   ├── launch_app.yaml
│   ├── launch_app_fresh.yaml
│   └── noop.yaml
├── scripts/
│   ├── run_all.sh            # Main orchestration + HTML report generator
│   ├── run_smoke.sh          # Smoke test subset
│   ├── run_failed.sh         # Re-run failed tests only
│   ├── tc_metadata.sh        # Test case metadata (descriptions, priorities, categories)
│   └── report_style.css      # External CSS for Jenkins-compatible reports
└── Jenkinsfile               # Pipeline definition
```

3.3 Key Design Decisions

- **Isolated test execution:** Each test clears app state and launches fresh, ensuring no cross-test contamination
- **Reusable helper flows:** Common sequences (onboarding, navigation) extracted into 9 shared helpers

- **Conditional recovery blocks:** `runFlow: when:` patterns handle transient UI states (loading screens, "Try again" dialogs, stylus bottom sheets)
- **Smart waits over fixed sleeps:** `extendedWaitUntil` with appropriate timeouts instead of arbitrary `sleep` calls
- **Video proof:** Each test execution is screen-recorded (local Jenkins runs)

4. BrowserStack Cloud Execution Results

4.1 Build Summary

Field	Value
Build ID	7824f1a16afa3c3968f2a80949a1ebeb6188486d
Device	Samsung Galaxy S24 (Android 14.0)
Tests Executed	35
Passed	32
Failed	3
Skipped	0
Timed Out	0
Pass Rate	91.4%

4.2 Detailed Results

Passed Tests (32/35)

#	Test Case	Duration	Module
1	TC01 - Language Screen Load	16s	Onboarding
2	TC02 - Language Selection & Navigation	39s	Onboarding
3	TC03 - Skip Name Onboarding	71s	Onboarding
4	TC04 - Enter Name Onboarding	50s	Onboarding
5	TC05 - Weather Widget & Location	55s	Home
6	TC06 - Type Question & AI Response	147s	Chat
7	TC08 - Home Feed Scroll	69s	Home
8	TC09 - Gender Selection Card	66s	Home

#	Test Case	Duration	Module
9	TC09 - Home Input Type	134s	Chat
10	TC10 - Home Weather Widget	126s	Home
11	TC11 - Listen AI Response (TTS)	114s	Chat
12	TC12 - Settings Display Mode	51s	Settings
13	TC13 - Settings Update Name	75s	Settings
14	TC14 - Settings Signup Phone	53s	Auth
15	TC15 - Settings Signup Skip	48s	Settings
16	TC16 - Drawer Signup Phone	49s	Auth
17	TC17 - Help FAQ Accordion	48s	Help
18	TC18 - Help Terms of Use	50s	Help
19	TC19 - Help Privacy Policy	63s	Help
20	TC20 - Drawer Change Language	62s	Navigation
21	TC21 - Speak Button UI	53s	Chat
22	TC23 - Login Phone OTP	75s	Auth
23	TC24 - Chat History Screen	114s	Chat
24	TC25 - Settings Logout	85s	Settings
25	TC27 - Location Interstitial Skip	61s	Navigation
26	TC28 - Chat Close Button	81s	Chat
27	TC29 - Legal Terms of Use	33s	Onboarding
28	TC30 - Legal Privacy Policy	26s	Onboarding
29	TC31 - Drawer Recent Chat	137s	Navigation
30	TC36 - Chat Follow-Up Question	138s	Chat
31	TC37 - Drawer Home Navigation	64s	Navigation
32	TC38 - Legal Dialog Close	35s	Onboarding

Average test duration: ~72 seconds

Failed Tests (3/35)

Test Case	Duration	Root Cause	Severity
TC07 - Home Screen Verification	80s	"Learn More" feed card not present in server response on real device	Low -- server content timing
TC10 - Share AI Response	105s	Share sheet text differs on Samsung Galaxy S24 vs emulator	Low -- device-specific UI
TC26 - Save AI Response	100s	Save confirmation toast disappeared before assertion	Low -- timing sensitivity

All 3 failures are environment/timing-related, not functional bugs in the app.

4.3 Tests Excluded from BrowserStack (5)

Test Case	Reason
TC32 - Photo Query Camera	Requires camera hardware simulation (not supported on BrowserStack)
TC33 - Error Screen Retry	Requires airplane mode toggle via ADB (not available on cloud)
TC34 - Chat Error Recovery	Requires network manipulation via ADB
TC35 - Chat History Retry	Requires network manipulation via ADB
TC09 - Photo Query Gallery	Requires pre-pushed gallery image via ADB

5. CI/CD Pipeline**5.1 Jenkins (Local)**

- **Automated pipeline** via `Jenkinsfile` with stages: Emulator Setup, App Install, Test Execution, Report Generation
- **HTML reports** with:
 - Executive summary with pass/fail/skip counts and progress bar
 - Per-test priority badges (P0/P1/P2) and module category tags
 - Embedded screen recording videos for each test case
 - Collapsible execution logs with color-coded step results
 - Environment details (device, OS, app version, Maestro version)
- **Content Security Policy** configured for inline styles, scripts, and video playback

5.2 BrowserStack (Cloud)

- **REST API integration** for app upload, test suite upload, and build triggering
- **Real device execution** on Samsung Galaxy S24 (Android 14)
- Tests packaged as self-contained zip with hardcoded configuration (no external environment dependency)
- Device logs and network logs enabled for debugging

6. Key Challenges & Solutions

Challenge	Solution
Emulator screen recordings were empty (0 bytes)	Switched from SIGTERM to SIGINT (<code>kill -2</code>) for graceful <code>screenrecord</code> shutdown
Jenkins CSP blocked inline styles and videos	Created persistent Groovy init script + external CSS file
BrowserStack env vars not propagating to sub-flows	Hardcoded all configuration values directly in YAML flows
BrowserStack test suite path resolution failures	Restructured zip with single parent folder as per BrowserStack docs
"Try out your stylus" overlay blocking input	Added conditional dismiss logic + ADB settings to disable stylus features
Server-side content changes (gender/crops cards)	Read-only assertions without clicking to preserve server state
Slow test execution on Jenkins	Replaced fixed sleeps with smart polling loops; batched ADB commands

7. Recommendations & Next Steps

1. **Fix 3 BrowserStack failures:** Make assertions device-agnostic (flexible text matchers, longer toast wait times)
2. **Multi-device testing:** Expand BrowserStack runs to additional devices (Pixel 8, OnePlus, different Android versions)
3. **Scheduled runs:** Set up daily/nightly automated test runs on BrowserStack
4. **App version gating:** Integrate test suite into the release pipeline to block releases with P0 test failures
5. **Flaky test monitoring:** Track pass rates over time to identify and address intermittent failures
6. **Accessibility testing:** Add accessibility checks using BrowserStack's accessibility scanning tools

8. Summary

The FarmerChat automation suite provides **comprehensive coverage of 40 test scenarios** across all critical user journeys. With a **91% pass rate on real Samsung Galaxy S24 hardware** via BrowserStack, the suite validates that the app's core functionality -- onboarding, AI chat, authentication, navigation, and settings -- works reliably for end users. The 3 failures are attributable to environment differences, not application defects.

The infrastructure supports both local Jenkins execution with rich HTML reporting (including video proof) and cloud-based real device testing via BrowserStack, giving the team confidence in every release.

Report generated: March 2026

Framework: Maestro | Cloud: BrowserStack App Automate | CI: Jenkins